Benchmark-Relative Systematic Portfolio Construction

Evidence from a Ten-ETF Historical Study

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Abstract

We study how tracking-error and turnover limits shape monthly long-only portfolios in a surviving ten-ETF panel. Momentum, low-volatility and reversal preferences are evaluated against a monthly 60/40 policy in development (2011–2017), validation (2018–2022) and holdout (2023–September 2026). Formation reserves introduced after pre-holdout execution failures were frozen before holdout evaluation. Turnover limits restrict preference expression, while the objective's risk penalty and other constraints leave the 8% and 12% tracking-error caps inactive. The 5% high-state cap binds infrequently, limiting the original regime comparison's ability to distinguish allocations. The primary model has negative active-return point estimates in development and validation; both block intervals include zero. Holdout annualized arithmetic active return is 1.23% with a descriptive 95% block interval [-1.55%, 3.90%]. Class attribution traces holdout gains to high-yield credit and gold, despite negative holdout mean IC. A post-holdout 3% high-state diagnostic binds more often but provides no conclusive return benefit. This retrospective study separates portfolio construction from investment skill. Public daily portfolio ledgers support table verification; rebuilding allocations requires the original data vintage or a labelled fresh download.

Keywords: Portfolio construction; tracking error; turnover; exchange-traded funds; walk-forward backtesting.

Code, protocol and evidence: PortfolioOS research repository.

1 Introduction and research questions

Benchmark-relative managers must translate investment preferences into holdings while controlling deviations from a policy portfolio and the cost of changing positions. A hard tracking-error limit can appear unimportant if a quadratic risk penalty already keeps the optimizer within that limit. Conversely, a turnover limit may prevent a desired allocation even when the risk budget has headroom. Separating these mechanisms matters when interpreting a flat constraint frontier or a small difference between fixed and state-dependent risk budgets.

We ask how risk and trading limits affect signal expression, whether greater expression improves net historical performance, and whether a tighter risk budget during high policy volatility changes allocations. The contribution is an auditable comparison in a transparent multi-asset ETF setting: declared constraint frontiers, independently costed comparison portfolios, preserved execution failures, and separate development, validation and holdout evidence. We measure constraint activity directly and reconcile active return by asset class, so construction effects can be distinguished from predictive skill. A separately labelled 3% high-state diagnostic was added after holdout review. The monthly 60/40 policy is a proposed comparison; the deterministic synthetic study remains a separate exercise.

2 Literature review

Roll's analysis of tracking error [1] establishes the distinction between optimization relative to a benchmark and conventional mean–variance efficiency. Jorion [2] derives the geometry of tracking-error-constrained portfolios and shows why controlling relative risk alone can leave total portfolio risk inefficient. Our objective penalizes active variance and also imposes a hard TE cap; realized total volatility and drawdown are reported alongside relative risk. We do not add a total-volatility mandate.

Clarke, de Silva and Thorley [3] extend the fundamental law of active management with a transfer coefficient describing how constraints impede the translation of forecasts into active holdings. Our *signal capture* diagnostic addresses that same implementation question, but is a score-expression ratio against a feasible reference, rather than their risk-adjusted cross-sectional correlation. Its inputs are preferences, not calibrated residual-return forecasts, so it is neither the classical transfer coefficient nor an estimate of expected information ratio.

Ledoit and Wolf motivate shrinkage covariance estimation [4]; DeMiguel, Garlappi and Uppal motivate the equal-weight comparison [5]. Künsch [6] provides the moving-block foundation for dependent-data resampling. Our fixed-length blocks wrap around the sample endpoints, following the circular procedure of Politis and Romano [7], rather than an unwrapped moving-block bootstrap. Momentum research documents winner–loser persistence in individual stocks [8]; volatility-return research studies stock-level aggregate and idiosyncratic risk [9]; reversal research considers short-horizon contrarian stock returns [10]. These findings motivate preferences, not calibrated ETF expected returns. Total ETF volatility is not idiosyncratic stock volatility, and a 21-session reversal score is not a replication of a weekly stock-reversal study. Economic portability is an empirical question.

3 Historical dataset and data quality

The ordered universe is SPY, IWM, EFA, EEM, IEF, TLT, LQD, HYG, GLD and VNQ. Equity covers the first four; fixed income covers IEF, TLT, LQD and HYG; gold and real estate are separate constraint groups. VNQ is treated as a separate real-estate allocation despite its equity-like risk. Funds were selected retrospectively and survived to retrieval: this is not an unbiased historical investable universe. Every fund

existed before January 2009; issuer inception and first-listing sources are recorded in the provenance appendix document. GLD's November 18, 2004 availability date is its first trading date, distinct from the trust's November 12 formation date.

The TigZig public API supplies Yahoo Finance data via yfinance. Its documented auto-adjusted Close incorporates splits and distributions [11]. The action ledger is preserved without a second adjustment. The operator's terms describe educational/demonstration use and disclaim accuracy [12]; an institutional licence and upstream redistribution permission are not claimed. Raw responses remain outside Git. Adjusted returns approximate total returns with provider reinvestment conventions; tax and exact distribution reinvestment execution are not modelled. Latest-vintage revisions can differ from historical information available at each date.

Validation checks hashes, strict timestamp order, duplicates, positive finite prices, common NYSE sessions, per-fund availability, distributions/splits, 20% absolute daily-return review thresholds, original-response Close values, and actions from a second endpoint of the same provider. The second endpoint is not an independent market-data validation.

The panel contains 4,463 common sessions from January 2, 2009 to September 30, 2026, with 0 missing sessions and 0 threshold exceedances. The ledger contains 1,133 distributions and no splits in this window; GLD has no cash distributions.

4 Mathematical model

With score s, benchmark b, annual covariance Σ and known lagged holdings w^{known}, the existing convex program solves

$$\max_{w} \; s^\top w - 10(w-b)^\top \Sigma (w-b) - 0.1\|w - w^{\text{known}}\|_1.$$

Fully invested long-only portfolios satisfy $\mathbf{1}^\top w = 1$ and $0 \le w_i \le 0.35$. At execution, annual estimated TE is at most 8% (5% in the high-state alternative), absolute active group exposure is at most 10 percentage points, and $T = \frac{1}{2}\|w - w^{\text{pretrade}}\|_1 \le 0.30$. Every accepted solution is independently checked. Infeasible or inaccurate solver outcomes stop a path rather than silently relaxing constraints.

At return date t, signal and decision information end at $t-2$; target weights are formed then, modelled at close $t-1$, and earn returns after that close. Execution-close drift changes actual turnover but does not change target formation. Holdings drift as

$$w_{i,t}^{\text{end}} = \frac{w_{i,t}(1+r_{i,t})}{1 + w_t^\top r_t}, \qquad R_t^{\text{net}} = w_t^\top r_t - \frac{c}{10000} T_t.$$

The additive fee convention is a declared NAV approximation; this does not prove closing prices were executable. Exact share/cash accounting is omitted.

5 Signals and covariance methodology

For information date u, raw momentum is $P_{u-21}/P_{u-252} - 1$; low volatility is the negative sample standard deviation of the last 63 daily returns; reversal is $-(P_u/P_{u-21} - 1)$. Cross-sectional 5th/95th winsorization and population-standard-deviation standardization precede the composite $s = z_{\text{momentum}} + z_{\text{lowvol}} + 0.5 z_{\text{reversal}}$. Coefficients are fixed preferences, not calibrated expected returns. With ten complete observations and linear quantile interpolation, 5th/95th winsorization clips only the minimum and maximum of each raw score, leaving eight values unchanged before standardization. This is limited tail treatment in a small cross-section, not strong protection against outliers; an isolated extreme can still be reduced

materially. Standardizing equities, bonds, gold and real estate together mixes distinct risk and return distributions and can encode asset-class preferences rather than within-class predictive information. Covariance uses 252 trailing returns, Ledoit–Wolf shrinkage, a 10^{-10} daily diagonal ridge and annualization 252. Sample covariance is a declared robustness alternative. Forward IC is Spearman correlation with subsequent asset returns through the next rebalance; the terminal period can be partial.

Signal capture is $(w - b)^\top s$ divided by the expression of a same-score, same-information long-only reference with the 35% position cap but without TE, group, turnover constraints or penalties. Exposure is measured against the realized execution benchmark for both paths. Absolute reference exposure at most 10^{-10} yields an undefined observation. Different ablations have different reference scores, so their capture ratios are not cardinal skill comparisons. High capture does not imply superior returns. This diagnostic is related to the transfer-coefficient literature [3], but its normalization is an attainable score exposure, not a risk-adjusted correlation.

6 Policy benchmark and constraints

The policy weights, in universe order, are $(0.30, 0.05, 0.15, 0.10, 0.20, 0.05, 0.15, 0, 0, 0)$. It resets at the close immediately before the first observed session return of each month and drifts between resets. The schedule is independently represented when the active strategy rebalances weekly. Ordinary policy, equal-weight and inverse-volatility comparisons satisfy their own rules and portfolio accounting; active-manager TE, sector and turnover mandates are not imposed on them. Both frictionless and independently cost-adjusted policy returns are exported. Each independent period starts endowed at its execution-date policy holdings; the active transition is charged and the initial policy endowment is not charged.

6.1 Pre-holdout design revision after execution failures

Original development/validation active paths without reserves failed execution checks, predominantly turnover limits following unobserved one-session drift; those failures remain archived. After inspecting these development/validation outcomes, we introduced identical formation reserves across active scenarios: 2 percentage points of turnover, 1 percentage point of group exposure and 0.1 percentage point of annual TE. Thus primary formation caps are 28%, 9 percentage points and 7.9%; the high-state formation TE cap is 4.9%. Execution mandates remain 30%, 10 percentage points and 8%/5%, respectively. This is an outcome-informed engineering correction to the initial specification. The corrected specification was reviewed and frozen before constructed holdout portfolio performance was examined; published results use that corrected model. Reserves address engineering feasibility and do not guarantee that execution constraints remain feasible. Remaining weekly sector failures in development and validation were retained. A benchmark target fits the 35% cap; transition constraints can still be infeasible.

7 Experiment design and holdout protocol

Warmup is January 2009–December 2010; development is January 3, 2011–December 29, 2017; validation is January 2, 2018–December 30, 2022; final holdout is January 3, 2023–September 30, 2026. Models receive physical price slices ending at each period boundary. Before holdout, data checks, tests, development/validation review and the configuration were frozen with dataset hashes and a code commit. An exclusive access ledger was written before the single final evaluation. No favourable ablation was substituted for the primary model. Inherited source discovery and issuer verification exposed incidental holdout-era quotes/performance panels, and full prices were ingested for quality checks; constructed

holdout performance was not used for selection. The historical split is retrospective, not prospective preregistration.

Experiments A–G cover TE caps 4/8/12%; turnover caps 10/30/50%; identical-trade cost sensitivities 0/10/25/40 bps; fixed 8% versus normal 8%/high 5% TE; three signal omissions plus regime awareness; the five portfolio strategies; and sample covariance, three-session information delay and weekly rebalancing. The full ledgers retain failures. Cost assumptions do not reoptimize holdings. Metrics use 252 observations/year and sample return standard deviations; Sharpe/Sortino assume zero risk-free return. Undefined metrics are explicit.

Uncertainty uses circular 21-session blocks, 2,000 resamples, seed 20261008 and 95% percentile intervals for annualized arithmetic active means [7]. Strategy differences are paired on the same dates. Approximate within-period stationarity and dependence within blocks are assumed; structural breaks, longer regime persistence and multiple comparisons are not resolved. No naive independent-day significance tests or persistent-alpha conclusions are made.

8 Empirical performance results

All returns below are computed from executed paths. CAGR, annualized arithmetic active return and TE are distinct quantities. Active returns use the frictionless policy; a cost-adjusted policy comparison is included in the machine-readable tables. Percent figures are expressed in percentage points.

8.1 Development

Table 1. Development-period performance.

Strategy	Gross CAGR	Net CAGR	TE	Active	Max. DD
Policy (net)	7.987	7.976	0.004	-0.011	-11.373
Equal weight	6.879	6.859	2.462	-1.082	-11.692
Inverse volatility	6.439	6.384	4.355	-1.697	-9.245
Fixed risk	7.722	7.499	3.653	-0.380	-17.784
Regime aware	7.710	7.486	3.626	-0.396	-17.440

Notes: All values are percentages. Active denotes annualized arithmetic return relative to the frictionless policy; TE is realized tracking error. Max. DD is maximum drawdown. Net returns include 10 bps per unit of one-way turnover.

Fixed-risk arithmetic active return: -0.38%; 95% block interval [-3.14%, 2.08%]. Mean capture: 0.665, with 84 valid and 0 excluded rebalance observations. Mean forward IC is 0.012. Frictionless policy CAGR is 7.99%; independently cost-adjusted policy CAGR is 7.98%.

8.2 Validation

Table 2. Validation-period performance.

Strategy	Gross CAGR	Net CAGR	TE	Active	Max. DD
Policy (net)	3.284	3.271	0.004	-0.013	-24.271
Equal weight	2.766	2.743	2.752	-0.618	-24.064
Inverse volatility	1.461	1.400	5.436	-2.228	-22.547
Fixed risk	2.283	2.033	3.294	-1.148	-25.315
Regime aware	2.330	2.086	3.250	-1.093	-25.315

Notes: All values are percentages. Active denotes annualized arithmetic return relative to the frictionless policy; TE is realized tracking error. Max. DD is maximum drawdown. Net returns include 10 bps per unit of one-way turnover.

Fixed-risk arithmetic active return: -1.15%; 95% block interval [-3.67%, 1.36%]. Mean capture: 0.548, with 60 valid and 0 excluded rebalance observations. Mean forward IC is -0.060. Frictionless policy CAGR is 3.28%; independently cost-adjusted policy CAGR is 3.27%.

8.3 Holdout

Table 3. Holdout-period performance.

Strategy	Gross CAGR	Net CAGR	TE	Active	Max. DD
Policy (net)	12.325	12.315	0.003	-0.009	-9.729
Equal weight	11.405	11.379	2.841	-0.853	-9.785
Inverse volatility	9.093	9.034	3.746	-3.160	-8.347
Fixed risk	13.980	13.717	2.782	1.233	-11.089
Regime aware	13.964	13.700	2.773	1.218	-11.089

Notes: All values are percentages. Active denotes annualized arithmetic return relative to the frictionless policy; TE is realized tracking error. Max. DD is maximum drawdown. Net returns include 10 bps per unit of one-way turnover.

Fixed-risk arithmetic active return: 1.23%; 95% block interval [-1.55%, 3.90%]. Mean capture: 0.730, with 45 valid and 0 excluded rebalance observations. Mean forward IC is -0.019. Frictionless policy CAGR is 12.33%; independently cost-adjusted policy CAGR is 12.32%.

9 Constraint sensitivity findings

Table 7 reports every declared scenario. Relaxing a risk cap need not change an allocation when another constraint or the objective dominates. The 8% and 12% TE configurations have essentially identical performance, because those caps are inactive at every recorded formation decision. The $\lambda = 10$ quadratic active-risk penalty already discourages large risk deviations, together with the group, position and turnover constraints; expanding the hard cap from 8% to 12% therefore adds headroom the optimizer does not use. The cap comparison alone does not isolate the risk penalty's marginal effect. Turnover restricted signal expression more than the tested 4%–12% TE range under this model, objective, universe and other constraints. This conditional sensitivity does not establish that turnover is generally more important than tracking error in portfolio management. Higher capture does not establish better returns. Capture and turnover frontiers describe construction mechanics rather than investment skill. Formation reserves apply identically across frontier scenarios; 10/30/50% turnover mandates imply 8/28/48% formation caps.

10 Volatility-regime comparison

The classifier uses the monthly-policy return history: trailing 63-session sample volatility times $\sqrt{252}$, compared with the expanding 75th percentile of strictly earlier valid rolling observations, requiring 252 earlier values. Only strict exceedances are high; ties are normal. Labels used in allocation are dated at the lagged information close and budgets change at rebalances. The timeline shows observation-date labels, whose availability is after that close. Conditional results use daily lagged labels and arithmetic summaries; they do not compound discontinuous subsequences or imply regime causality.

Development: regime-aware minus fixed annualized arithmetic return -0.02%; paired 95% block interval [-0.12%, 0.10%].

Validation: regime-aware minus fixed annualized arithmetic return 0.05%; paired 95% block interval [-0.07%, 0.18%].

Holdout: regime-aware minus fixed annualized arithmetic return -0.02%; paired 95% block interval [-0.10%, 0.05%].

Table 4. Conditional performance by lagged volatility state.

Period	Strategy	State	Days	Active	TE
Development	Fixed risk	normal	1607	-0.561	3.241
Development	Fixed risk	high	154	1.507	6.577
Development	Regime aware	normal	1607	-0.581	3.246
Development	Regime aware	high	154	1.532	6.372
Validation	Fixed risk	normal	810	-0.755	2.786
Validation	Fixed risk	high	449	-1.855	4.056
Validation	Regime aware	normal	810	-0.686	2.767
Validation	Regime aware	high	449	-1.827	3.979
Holdout	Fixed risk	normal	722	1.506	2.545
Holdout	Fixed risk	high	217	0.328	3.461
Holdout	Regime aware	normal	722	1.512	2.541
Holdout	Regime aware	high	217	0.241	3.439

10.1 Constraint activity and a binding-cap diagnostic

The original regime comparison is partly mechanical. Fixed-risk realized TE is about 2.8%–3.7%, below the 5% high-state mandate, although realized TE alone cannot determine whether an ex-ante limit binds. At formation, after the 0.1-percentage-point reserve, the 4.9% cap binds at only 2/7, 5/21 and 2/10 high-state rebalances in development, validation and holdout. The budgets differ only on high-state rebalance dates, and the risk penalty already limits many desired deviations. Binding means estimated formation TE is within 10^{-5} of the formation cap (0.1 basis point of annual TE); this is a numerical activity diagnostic, not a dual-value or realized-risk test. We therefore add one separately labelled post-holdout diagnostic with 8% normal and 3% high-state mandates, keeping all other settings fixed. The 2.9% high-state formation cap binds appreciably more often. This experiment was requested after holdout review and is exploratory; it does not alter the frozen specification or its reported performance.

Table 5. Post-holdout 8%/3% regime diagnostic.

Period	High binding	Net CAGR	Active	TE	Δ vs fixed
Development	5/7	7.822	-0.086	3.501	0.294
Validation	11/21	2.279	-0.913	3.100	0.234
Holdout	7/10	13.334	0.886	2.578	-0.347

Notes: Returns and TE are percentages. High binding is binding/high-state rebalance counts. Δ is annualized mean daily net-return difference from fixed 8% risk, in percentage points/year.

Development paired 95% circular-block interval for the difference: [-0.228, 0.979] percentage points/year. Validation paired 95% circular-block interval for the difference: [-0.238, 0.712] percentage points/year. Holdout paired 95% circular-block interval for the difference: [-1.025, 0.250] percentage points/year. All paired intervals include zero. A binding cap changes allocations and reduces realized TE, but this diagnostic does not establish a return benefit from the classifier.

Only 38 high-state decision dates distinguish budgets (7, 21 and 10; 23 binding at 3%). This regime test is underpowered: the narrow original paired intervals reflect similar return paths and few decisions, not strong evidence about a general regime effect.

11 Robustness and ablation results

The two low-volatility-omission TE entries in Table 7 both round to 4.989%. Recomputing from distinct daily return ledgers gives 4.988843319% (development) and 4.988632707% (validation). Independent pre-holdout reruns reproduce both paths within solver tolerances; this is a rounding coincidence, not a copied value. The alternative covariance, delayed-information and signal-omission experiments remain separately labelled. They are descriptive checks, not an uncontrolled parameter search. Weekly paths that breach execution constraints receive no full-period CAGR or fabricated continuation. The omission plots preserve the full original model rather than selecting the best historical omission.

Weekly has the highest declared holdout net CAGR (14.99%) and 2.38% arithmetic active return. Both earlier paths failed execution checks, leaving no full-period development or validation evidence. The tall holdout bar in Figure 8 therefore does not validate a repeatable weekly advantage.

12 Economic interpretation

Executed holdings explain the active-return contributions below, with trading costs deducted separately. Standardized preference scores need not predict portfolio returns.

Table 6. Fixed-risk arithmetic active-return attribution.

Component	Development	Validation	Holdout
Equity	0.362	-0.981	-1.684
Fixed income	-0.697	-0.195	2.014
Gold	0.149	0.124	1.072
Real estate	0.014	0.149	0.063
Trading costs	-0.207	-0.245	-0.231
Net active return	-0.380	-1.148	1.233

Notes: Percentage points/year, computed as 252 times mean daily $\sum_{i \in g}(w_{i,t} - b_{i,t})r_{i,t}$. Costs are a separate negative component. These are additive arithmetic contributions, not a CAGR decomposition.

In holdout, equity exposure averages 5.96 percentage points below the policy and detracts 1.68 percentage points/year. SPY and IWM contribute -2.33 and -1.20.

The holdout gain is largely long high-yield credit and gold during 2023–September 2026. The policy assigns zero weight to both HYG and GLD; their contributions are +2.21 and +1.07 percentage points/year, totaling +3.28 against 1.23 net active return. Losses elsewhere and trading costs absorb the difference. GLD's 124.5% cumulative adjusted return over the executed holdout sessions is a likely driver of its contribution; HYG is the larger arithmetic contributor. [1]

Mean forward IC ranks all ten assets, whereas the positive active P&L is concentrated in these two positions. That concentration, together with the different weighting and horizons of rank IC and daily holdings-based returns, explains why negative average IC can coexist with positive active return. These window-specific credit and gold exposures do not establish broad predictive skill or risk-adjusted alpha. No regression-based risk-adjusted alpha, causal regime effect or investable closing-auction advantage is established.

13 Limitations

The surviving universe, latest adjustment vintage and Yahoo-derived data are material selection and measurement limitations. ETF prices embed fund expenses; taxes, share-lot/cash execution, market impact, liquidity, funding and settlement are omitted. Turnover is half-L1 and fees are additive fractions of beginning NAV. Constraints hold at accepted rebalances; drift can violate limits between them, and realized TE can exceed ex-ante caps. Fixed risk reserves cannot guarantee future mandate compliance. The modest fixed panel and multiple comparisons limit generalization; block intervals do not solve structural-break uncertainty. Zero-rate Sharpe is not a comparison with historical cash yields. The split is retrospective and independently reproduced source prices can change vintage. The original market-data bundle is retained locally and is not publicly available. Fingerprints establish identity but cannot supply those inputs; exact reconstruction of allocations is therefore conditional on original-bundle access. Public daily portfolio returns and decision ledgers support independent recomputation of the performance tables without provider data. Fresh downloads can support method replication while changing numerical findings.

14 Conclusion

Under the tested objective and mandates, turnover restricts signal expression while the 8% and 12% TE caps remain inactive. Holdout active return concentrates in HYG and GLD, despite negative holdout mean IC. Formation reserves were an outcome-informed correction frozen before holdout. The regime comparison offers limited decision evidence and inconclusive return differences. These findings describe construction mechanics and window-specific exposures; they establish neither a general constraint ranking nor persistent alpha.

[1] Compounds all 939 daily returns dated January 3, 2023–September 30, 2026, including the first return from the December 30, 2022 close; see the endpoint ledger in Appendix B.

Research figures

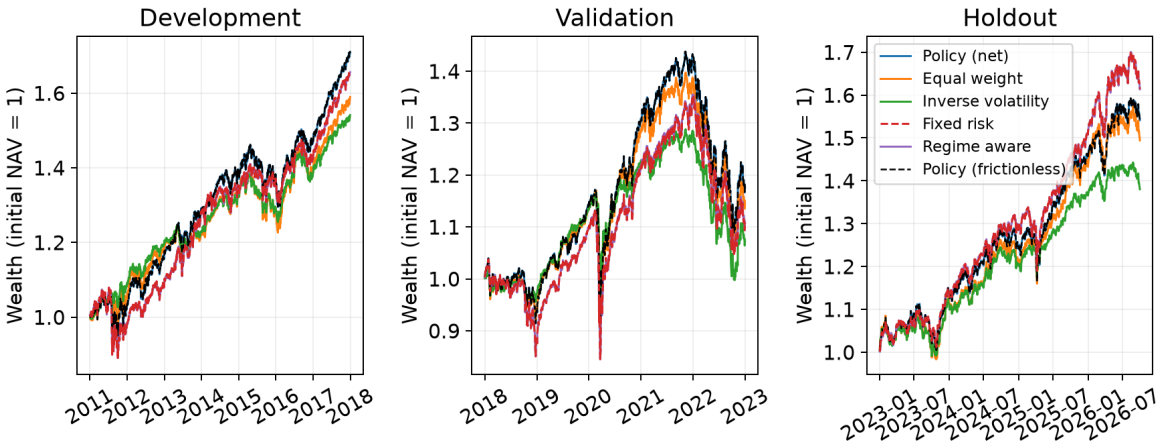


Figure 1. Executed cumulative net wealth with the frictionless policy shown separately. Each period is independently endowed at NAV 1.

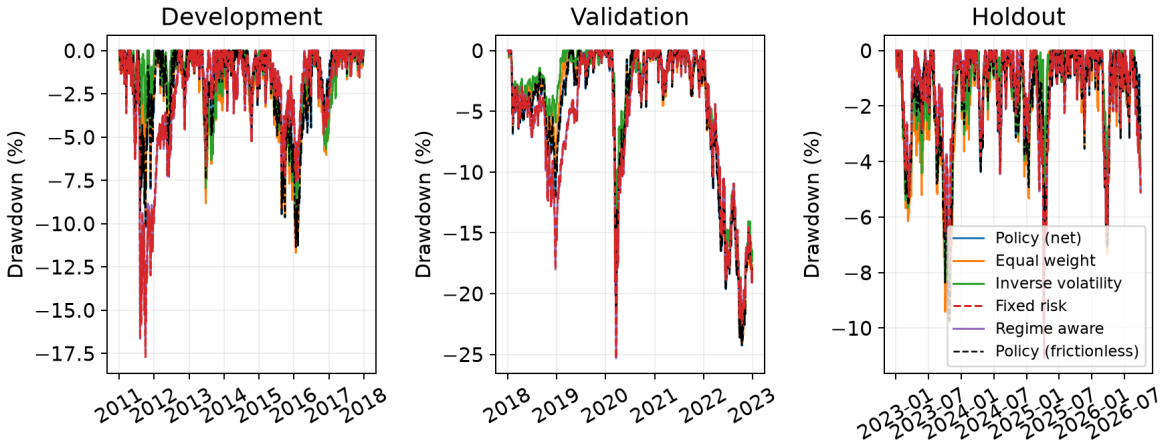


Figure 2. Executed net drawdowns, including initial NAV 1 in the running high-water mark.

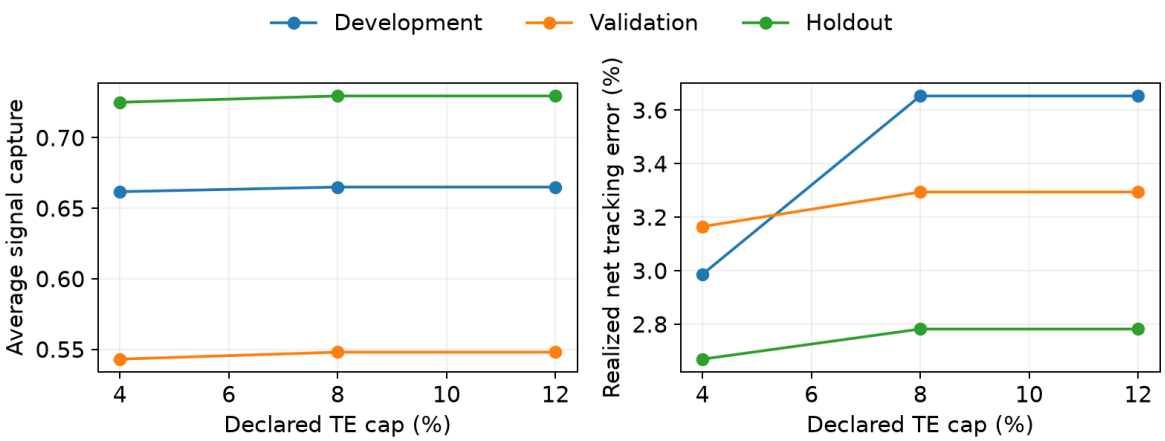


Figure 3. Declared TE mandates versus average signal capture and realized net TE; all other primary settings fixed.

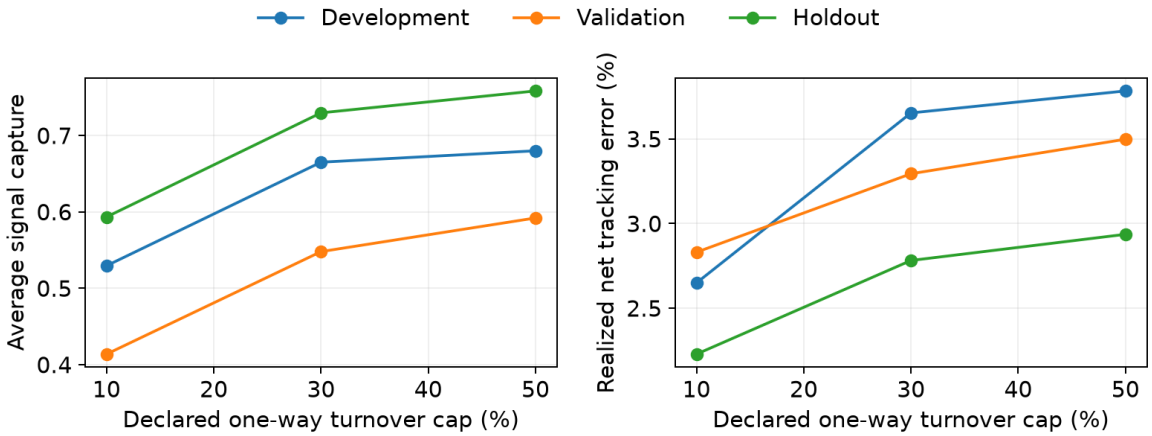


Figure 4. Declared turnover mandates versus capture and realized net TE; the identical two-percentage-point formation reserve applies throughout.

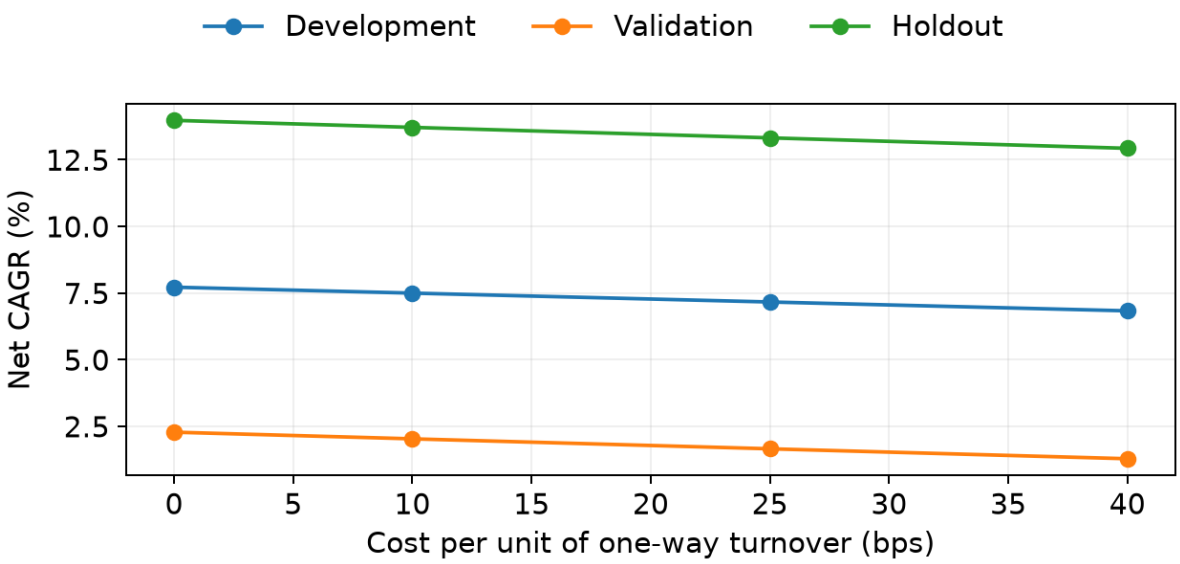


Figure 5. Net CAGR under accounting-only cost changes with identical gross returns, holdings and trades.

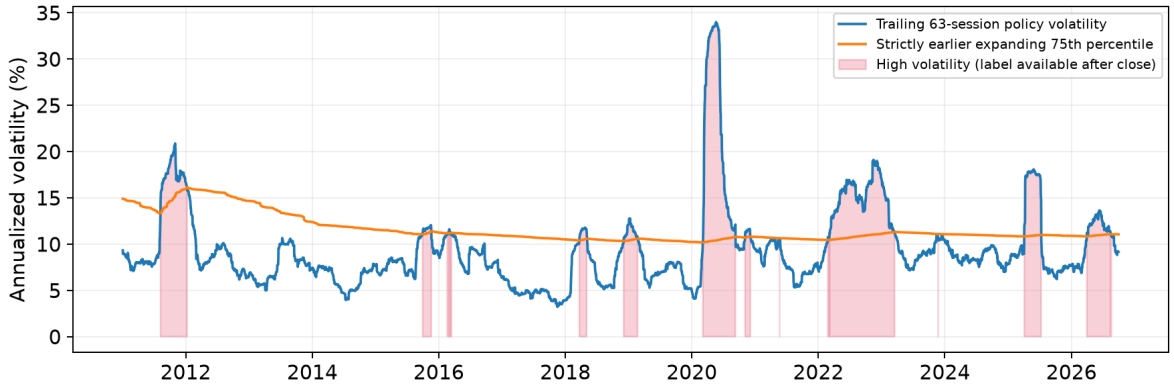


Figure 6. Policy volatility and its strictly earlier expanding threshold. Red shading marks observation-date high-state labels, available only after that close.

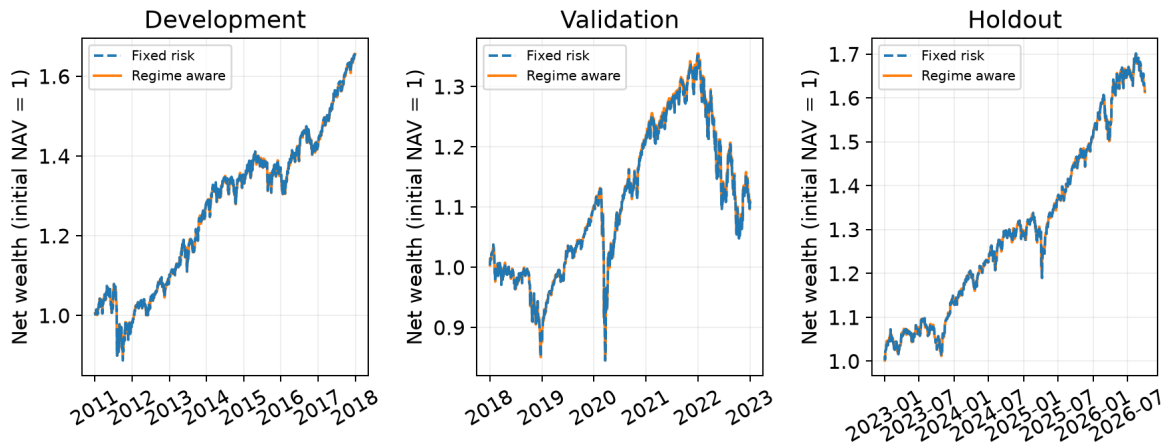


Figure 7. Executed fixed-risk and regime-aware net wealth; small visual differences need not indicate meaningful skill.

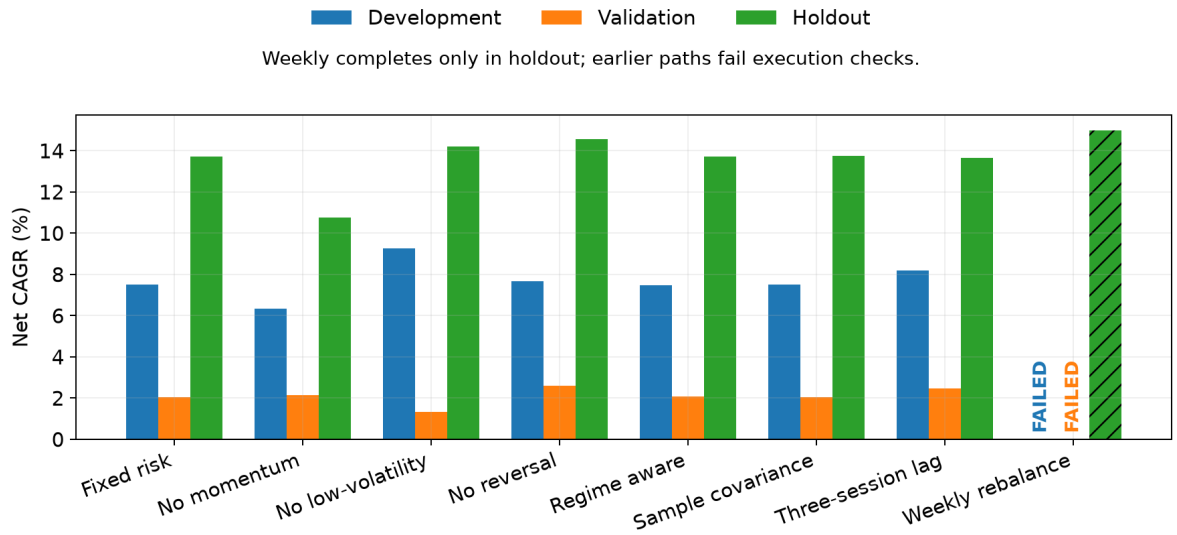


Figure 8. Executed ablations and robustness comparisons. FAILED denotes an interrupted path with no full-period return bar. The hatched weekly holdout bar has no completed development or validation counterpart.

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A All declared scenario results

Table values are percentages for CAGR, active return, TE and turnover; capture is unitless. A dash denotes an unavailable full-period metric. Complete Sharpe, Sortino, volatility, drawdown, information ratio, wealth, cost drag, active share, IC and constraint diagnostics are in `all_performance.csv` and the per-scenario ledgers.

Table 7. All executed scenarios; failures are retained.

Period	Scenario	Net CAGR	Active	TE	Capture	Turnover
Development	Fixed risk	7.499	-0.380	3.653	0.665	17.210
Development	TE 4%	7.790	-0.168	2.986	0.662	17.273
Development	TE 8%	7.499	-0.380	3.653	0.665	17.210
Development	TE 12%	7.499	-0.380	3.653	0.665	17.210
Development	Turnover 10%	8.003	0.025	2.649	0.529	7.272
Development	Turnover 30%	7.499	-0.380	3.653	0.665	17.210
Development	Turnover 50%	7.532	-0.345	3.783	0.680	20.522
Development	Cost 0 bps	7.722	-0.173	3.651	0.665	17.210
Development	Cost 10 bps	7.499	-0.380	3.653	0.665	17.210
Development	Cost 25 bps	7.166	-0.690	3.658	0.665	17.210
Development	Cost 40 bps	6.834	-1.001	3.666	0.665	17.210
Development	Regime aware	7.486	-0.396	3.626	0.665	17.316
Development	No momentum	6.351	-1.581	2.386	0.419	17.198
Development	No low-volatility	9.256	1.402	4.989	0.629	21.105
Development	No reversal	7.678	-0.293	2.837	0.689	10.438
Development	Policy (net)	7.976	-0.011	0.004	–	0.885
Development	Equal weight	6.859	-1.082	2.462	–	1.557
Development	Inverse volatility	6.384	-1.697	4.355	–	4.241
Development	Sample covariance	7.494	-0.384	3.653	0.665	17.245
Development	Three-session lag	8.178	0.163	2.963	0.653	17.595
Development	Weekly rebalance	–	–	–	–	–
Validation	Fixed risk	2.033	-1.148	3.294	0.548	20.398
Validation	TE 4%	2.203	-0.979	3.165	0.543	19.746
Validation	TE 8%	2.033	-1.148	3.294	0.548	20.398
Validation	TE 12%	2.033	-1.148	3.294	0.548	20.398
Validation	Turnover 10%	2.451	-0.718	2.831	0.414	7.816
Validation	Turnover 30%	2.033	-1.148	3.294	0.548	20.398
Validation	Turnover 50%	2.692	-0.502	3.498	0.592	27.121
Validation	Cost 0 bps	2.283	-0.903	3.291	0.548	20.398
Validation	Cost 10 bps	2.033	-1.148	3.294	0.548	20.398
Validation	Cost 25 bps	1.659	-1.515	3.301	0.548	20.398
Validation	Cost 40 bps	1.286	-1.883	3.313	0.548	20.398
Validation	Regime aware	2.086	-1.093	3.250	0.546	19.938
Validation	No momentum	2.143	-1.108	3.547	0.385	19.809
Validation	No low-volatility	1.312	-1.765	4.989	0.576	20.635
Validation	No reversal	2.591	-0.642	3.208	0.604	12.233
Validation	Policy (net)	3.271	-0.013	0.004	–	1.112
Validation	Equal weight	2.743	-0.618	2.752	–	1.881
Validation	Inverse volatility	1.400	-2.228	5.436	–	5.065
Validation	Sample covariance	2.032	-1.148	3.293	0.548	20.375
Validation	Three-session lag	2.463	-0.745	3.276	0.534	21.237
Validation	Weekly rebalance	–	–	–	–	–

Table 7. All executed scenarios (continued).

Period	Scenario	Net CAGR	Active	TE	Capture	Turnover
Holdout	Fixed risk	13.717	1.233	2.782	0.730	19.135
Holdout	TE 4%	13.231	0.786	2.669	0.725	18.822
Holdout	TE 8%	13.717	1.233	2.782	0.730	19.135
Holdout	TE 12%	13.717	1.233	2.782	0.730	19.135
Holdout	Turnover 10%	14.216	1.639	2.228	0.593	7.338
Holdout	Turnover 30%	13.717	1.233	2.782	0.730	19.135
Holdout	Turnover 50%	13.508	1.040	2.936	0.758	22.762
Holdout	Cost 0 bps	13.980	1.465	2.782	0.730	19.135
Holdout	Cost 10 bps	13.717	1.233	2.782	0.730	19.135
Holdout	Cost 25 bps	13.323	0.887	2.785	0.730	19.135
Holdout	Cost 40 bps	12.931	0.540	2.792	0.730	19.135
Holdout	Regime aware	13.700	1.218	2.773	0.729	19.198
Holdout	No momentum	10.752	-1.481	2.518	0.423	16.535
Holdout	No low-volatility	14.202	1.891	4.250	0.594	20.959
Holdout	No reversal	14.577	2.005	2.951	0.807	10.956
Holdout	Policy (net)	12.315	-0.009	0.003	–	0.762
Holdout	Equal weight	11.379	-0.853	2.841	–	1.886
Holdout	Inverse volatility	9.034	-3.160	3.746	–	4.468
Holdout	Sample covariance	13.762	1.273	2.785	0.730	19.142
Holdout	Three-session lag	13.640	1.140	2.777	0.738	19.189
Holdout	Weekly rebalance	14.992	2.377	3.207	0.747	10.037

Scenario labels correspond to the identifiers in the declared protocol and `all_performance.csv`. The complete model is retained in every signal-omission comparison; no alternative is promoted to the primary specification.

Failure: development, weekly: Execution mandate failed on 2012-06-04: sector active weight, sector active weight.

Failure: validation, weekly: Execution mandate failed on 2020-03-16: sector active weight.

B Reproduction appendix

Frozen code commit: `86a72b72584cf70e990525b62a7d4fab6e3649bd`.

Dataset SHA-256:

`5af0b0c65a9b4f8be569090f974c0b5446839a0bac20c3485d2184109cf981ea`.

See `docs/empirical_phase2.md` for the data-provider and issuer sources, environment versions, original failures, exact commands and verification record.

Exact-input reproduction. The original bundle (original responses, prices, actions and manifest) is local and not publicly distributed. Exact-input reproduction requires access to that bundle, matching recorded fingerprints, the frozen model commit and protocol, and compatible recorded dependencies. Numerical agreement is subject to solver tolerances; byte-for-byte equality across environments is not promised. The local validation reproduction and accounting audit used the original bundle; they do not establish that an external reader can retrieve the same vintage. Dataset hashes identify the inputs but cannot provide them.

Approximate replication with a new vintage. The acquisition commands retrieve fresh provider data, whose historical adjusted prices or actions may have changed. This replicates the method, not necessarily the published numerical results. Preserve a new manifest and fingerprints and use a new explicitly labelled output directory. The original frozen holdout output must not be overwritten or retuned.

Table verification without provider data. The public evidence includes daily portfolio gross/net returns, frictionless and cost-adjusted policy returns, turnover/cost accounting, optimization diagnostics, capture ledgers and forward-IC ledgers. Use the verifier:

`scripts/verify_public_empirical_tables.py`

It recomputes CAGR, active means, TE, drawdown, turnover, capture and block intervals from those outputs. The separate `results/empirical_review_response` directory contains reconciled class attribution, full-precision Table 7 checks, cap-activity counts and the exploratory 3% high-state paths. The `holdout_position_endpoints.csv` ledger records normalized adjusted-price endpoints for HYG and GLD. Its hash, exact return window, preceding-close baseline and observation count are recorded in `holdout_position_context.json`. The public verifier checks endpoint arithmetic and dates. Supplying `--bundle` additionally recomputes the compounded daily return from the original adjusted series after verifying its frozen fingerprint. The directory also preserves hashes of the original evaluation evidence. These artifacts verify table arithmetic, not input-price accuracy or independent optimizer reconstruction. Individual ETF daily return series are not redistributed because upstream redistribution rights remain unverified. The reporting script reads executed ledgers and never calls the optimizer. The manuscript compiles from the repository root using the documented Tectonic command.